SNOWFLAKE

1. CREATION OF SCHEMA

```
USE WAREHOUSE TRAINING_WH;

USE DATABASE TRAINING_DB;


CREATE SCHEMA IF NOT EXISTS LAHARI_EDUCATION_SCHEMA;

USE SCHEMA LAHARI_EDUCATION_SCHEMA;

SHOW SCHEMAS IN DATABASE TRAINING_DB;
```

ults (just now)

able | Chart | Pivot

Sort ascending

4 rows | 53ms

created_on	name	is_default	is_current	database_name	owner	comment
2026-11-04 03:04:28.400 -0800	INFORMATION_SCHEMA	N	N	TRAINING_DB		Views descr
2026-11-04 03:03:42.110 -0800	LAHARI_EDUCATION_SCHEMA	N	Y	TRAINING_DB	ACCOUNTADMIN	
2026-11-02 02:12:36.776 -0800	PUBLIC	N	N	TRAINING_DB	ACCOUNTADMIN	
2026-11-02 02:14:08.134 -0800	RAW	N	N	TRAINING_DB	ACCOUNTADMIN	

2. FILE FORMAT CREATION

```
17  CREATE FILE FORMAT IF NOT EXISTS LAHARI_CSV_FORMAT
18      TYPE = CSV
19      SKIP_HEADER = 1
20      FIELD_OPTIONALLY_ENCLOSED_BY = '"'
21      EMPTY_FIELD_AS_NULL = TRUE;
22  Ctrl+I to generate
```

esults (just now)

Table | Chart | Pivot

	status
1	File format LAHARI_CSV_FORMAT successfully created.

3. STAGE CREATION AND UPLOADING FILES INTO STAGE

```
CREATE STAGE IF NOT EXISTS LAHARI_EDUCATION_STAGE
FILE_FORMAT = LAHARI_CSV_FORMAT;

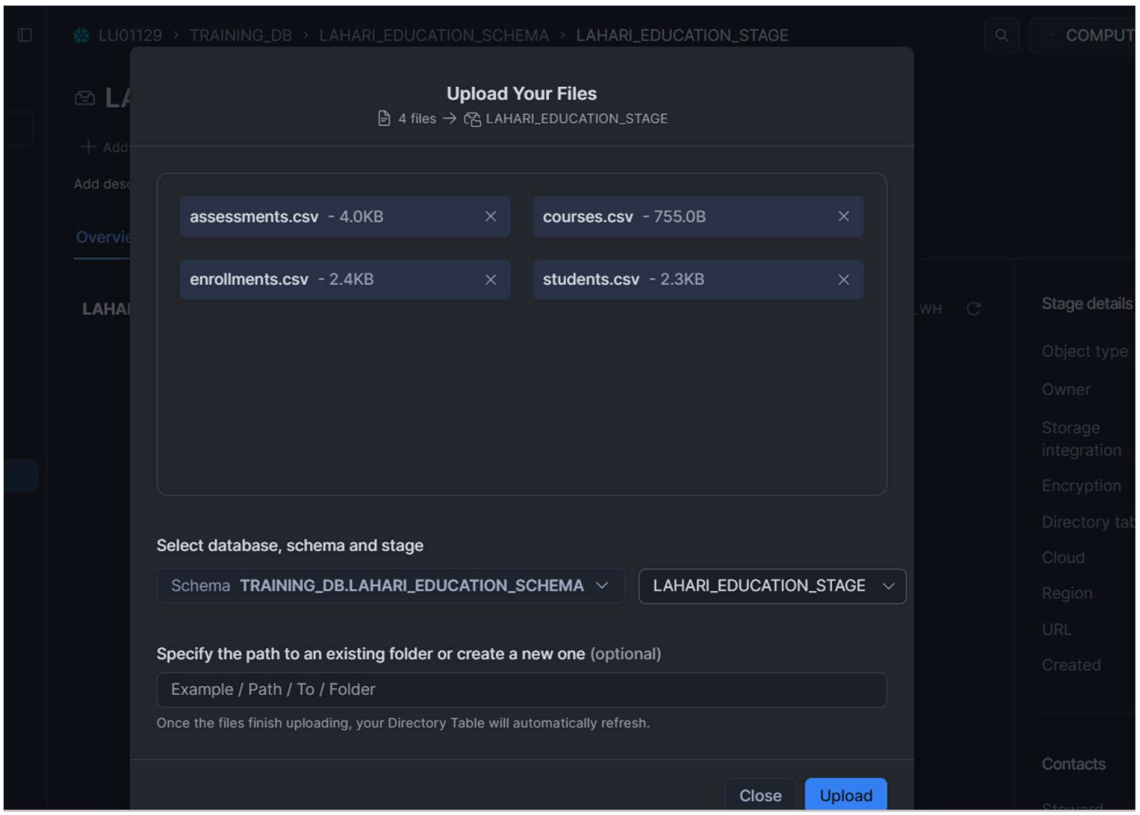
SHOW STAGES LIKE 'LAHARI_EDUCATION_STAGE';

 Ctrl+I to generate
```

sults (just now)

Table | Chart | Pivot — 1 row — 35ms

created_on	name	database_name	schema_name	url	has_credentials	has_encryp
2026-11-04 03:41:02	LAHARI_EDUCATION_S	TRAINING_DB	LAHARI_EDUCATION_S		N	N




```
LIST @LAHARI_EDUCATION_STAGE;
```

sults (just now)

Table | Chart | Pivot

name	size	md5
lahari_education_stage/assessments.csv	4080	3ca5568dd83856b65ab1b652429fed07
lahari_education_stage/courses.csv	768	af74e1a65274a2911a73d59871b9f91f
lahari_education_stage/enrollments.csv	2512	f5e05440a12fc39411125f72a93bd96e
lahari_education_stage/students.csv	2368	0fc7db1b9c6ef137da443aa4eb2b090f

4. TABLES CREATION AND COPYING DATA FROM STAGE INTO TABLES

```sql
CREATE TABLE IF NOT EXISTS LAHARI_STUDENTS (
    STUDENT_ID VARCHAR(20),
    STUDENT_NAME VARCHAR(100),
    EMAIL VARCHAR(150),
    DEPARTMENT VARCHAR(100),
    ADMISSION_DATE VARCHAR(30),
    CITY VARCHAR(50)
);

CREATE TABLE IF NOT EXISTS LAHARI_COURSES (
    COURSE_ID VARCHAR(20),
    COURSE_NAME VARCHAR(150),
    DEPARTMENT VARCHAR(100),
    INSTRUCTOR_NAME VARCHAR(100),
    COURSE_DURATION VARCHAR(20),
    COURSE_FEE VARCHAR(30)
);

CREATE TABLE IF NOT EXISTS LAHARI_ENROLLMENTS (
    ENROLLMENT_ID VARCHAR(20),
    STUDENT_ID VARCHAR(20),
    COURSE_ID VARCHAR(20),
    ENROLLMENT_DATE VARCHAR(30),
    COMPLETION_STATUS VARCHAR(30),
    PROGRESS_PERCENTAGE VARCHAR(20)
);
```

```sql
SHOW TABLES IN DATABASE TRAINING_DB;
```

esults (just now)

Table | Chart | Pivot | Sort ascending | 5 rows | 149ms

	created_on	name	database_name	schema_name	kind	comment	cluster_by
1	2026-11-04 03:48:41.284 -0800	LAHARI_ASSESSMENTS	TRAINING_DB	LAHARI_EDUCATION_SCHEMA	TABLE		
2	2026-11-04 03:48:08.544 -0800	LAHARI_COURSES	TRAINING_DB	LAHARI_EDUCATION_SCHEMA	TABLE		
3	2026-11-04 03:48:23.426 -0800	LAHARI_ENROLLMENTS	TRAINING_DB	LAHARI_EDUCATION_SCHEMA	TABLE		
4	2026-11-04 03:47:51.865 -0800	LAHARI_STUDENTS	TRAINING_DB	LAHARI_EDUCATION_SCHEMA	TABLE		
5	2026-11-02 02:20:25.491 -0800	EMPLOYEES	TRAINING_DB	RAW	TABLE		

```sql
COPY INTO LAHARI_STUDENTS
FROM @LAHARI_EDUCATION_STAGE/students.csv
FILE_FORMAT = (FORMAT_NAME = LAHARI_CSV_FORMAT)
ON_ERROR = 'ABORT_STATEMENT';

COPY INTO LAHARI_COURSES
FROM @LAHARI_EDUCATION_STAGE/courses.csv
FILE_FORMAT = (FORMAT_NAME = LAHARI_CSV_FORMAT)
ON_ERROR = 'ABORT_STATEMENT';

COPY INTO LAHARI_ENROLLMENTS
FROM @LAHARI_EDUCATION_STAGE/enrollments.csv
FILE_FORMAT = (FORMAT_NAME = LAHARI_CSV_FORMAT)
ON_ERROR = 'ABORT_STATEMENT';

COPY INTO LAHARI_ASSESSMENTS
FROM @LAHARI_EDUCATION_STAGE/assessments.csv
FILE_FORMAT = (FORMAT_NAME = LAHARI_CSV_FORMAT)
ON_ERROR = 'ABORT_STATEMENT';
```

Add to Chat | Explain | Quick edit | Format

ults (just now)

able | Chart | Pivot | 1 row

file	status	rows_parsed	rows_loaded	error_limit	errors_seen	first_error
lahari_education_stage	LOADED	100	100	1	0	null

5. VERIFYING TABLES

```
SELECT COUNT(*) AS TOTAL_STUDENTS
FROM LAHARI_STUDENTS;
```

sults (just now)

Table | Chart | Pivot

	# TOTAL_STUDENTS
1	30

```
SELECT COUNT(*) AS TOTAL_COURSES
FROM LAHARI_COURSES;
```

esults (just now)

Table | Chart | Pivot

	# TOTAL_COURSES
1	10

```
SELECT COUNT(*) AS TOTAL_ENROLLMENTS
FROM LAHARI_ENROLLMENTS;
```

Results (just now)

Table | Chart | Pivot

	# TOTAL_ENROLLMENTS
1	61

```
SELECT COUNT(*) AS TOTAL_ASSESSMENTS
FROM LAHARI_ASSESSMENTS;
```

esults (just now)

Table | Chart | Pivot

	# TOTAL_ASSESSMENTS
1	100

Table | Chart | Pivot — 10 rows — 372ms

	STUDENT_ID	STUDENT_NAME	EMAIL	DEPARTMENT	ADMISSION_DATE	CITY
	1001 10.0% 1002 10.0% +8 more	Aadhya Menon 10.0% Aarav Sharma 10.0% +8 more	aadhya.menon@example.com 10.0% aarav.sharma@example.com 10.0% +7 more	Computer Science 20.0% Data Science 20.0% +4 more	02-Jul-2024 10.0% 15/06/2024 10.0% +8 more	Chennai 20.0% Kochi 20.0% +6 more
1	1001	Aarav Sharma	aarav.sharma@example.com	Computer Science	2024-06-15	Chennai
2	1002	Diya Patel	diya.patel@example.com	Information Technology	15/06/2024	Mumbai
3	1003	Ishaan Rao	ishaan.rao@example.com	Data Science	2024/06/18	Bengaluru
4	1004	Meera Nair	null	Computer Science	20-Jun-2024	Kochi
5	1005	Kabir Singh	kabir.singh@example.com	Electronics	2024-06-22	Delhi
6	1006	Ananya Iyer	ananya.iyer@example.com	computer science	2024-06-25	Chennai
7	1007	Reyansh Gupta	reyansh.gupta@example.com	Information Technology	26/06/2024	Pune
8	1008	Saanvi Reddy	saanvi.reddy@example.com	Data Science	2024-06-28	Hyderabad
9	1009	Vivaan Das	vivaan.das@example.com	Electronics	2024/07/01	Kolkata
10	1010	Aadhya Menon	aadhya.menon@example.com	Business Analytics	02-Jul-2024	Kochi

Table | Chart | Pivot — 10 rows — 491ms

	COURSE_ID	COURSE_NAME	DEPARTMENT	INSTRUCTOR_NAME	COURSE_DURATION	COURSE_FEE
	201 10.0% 202 10.0% +8 more	Business Intelligence 10.0% Cloud Fundamentals 10.0% +8 more	Computer Science 30.0% Business Analytics 20.0% +4 more	Dr. Meenal Joshi 10.0% Dr. Kavitha Rao 10.0% +8 more	10 20.0% 12 20.0% +5 more	12000.0 10.0% 5500.0 10.0% +8 more
1	201	Python Programming	Computer Science	Dr. Kavitha Rao	12	8500.0
2	202	Database Management	Information Technology	Prof. Arvind Kumar	10	7800.0
3	203	Machine Learning	Data Science	Dr. Nisha Menon	14	12000.0
4	204	Cloud Fundamentals	information technology	Prof. Rahul Shah	8	6500.0
5	205	Data Visualization	Business Analytics	Dr. Sneha Iyer	6	5500.0
6	206	Cybersecurity Basics	Computer Science	Prof. Vikram Das	10	9000.0
7	207	Statistics for Analytics	Data Science	Dr. Pooja Nair	8	7000.0
8	208	Digital Electronics	Electronics	Prof. Sanjay Mehta	12	8200.0
9	209	Business Intelligence	Business Analytics	Dr. Meenal Joshi	9	9500.0
10	210	Web Development	Computer Science	Prof. Kiran Reddy	11	8800.0

Table | Chart | Pivot — 10 rows — 104ms

	ENROLLMENT_ID	STUDENT_ID	COURSE_ID	ENROLLMENT_DATE	COMPLETION_STATUS	PROGRESS_PERCENTAGE
	3001 10.0% 3002 10.0% +8 more	1001 20.0% 1002 20.0% +3 more	202 20.0% 203 20.0% +5 more	04/08/2024 10.0% 07-Aug-2024 10.0% +8 more	IN_PROGRESS 40.0% COMPLETED 30.0% +2 more	65 40.0% 100 30.0% +2 more
1	3001	1001	201	2024-08-01	COMPLETED	100
2	3002	1001	203	08/08/2024	IN_PROGRESS	65
3	3003	1002	202	04/08/2024	IN_PROGRESS	65
4	3004	1002	206	11-Aug-2024	NOT_STARTED	0
5	3005	1003	203	07-Aug-2024	completed	0
6	3006	1003	209	2024-08-14	COMPLETED	100
7	3007	1004	204	2024-08-10	COMPLETED	100
8	3008	1004	202	17/08/2024	IN_PROGRESS	65
9	3009	1005	205	13/08/2024	IN_PROGRESS	65
10	3010	1005	205	20-Aug-2024	NOT_STARTED	115

Table | Chart | Pivot — 10 rows — 364ms

	ASSESSMENT_ID	ENROLLMENT_ID	ASSESSMENT_NAME	ASSESSMENT_DATE	MARKS_OBTAINED	TOTAL_MARKS
	5001 10.0% 5002 10.0% +8 more	3002 20.0% 3003 20.0% +4 more	Midterm 60.0% Final Assessment 40.0%	03/09/2024 10.0% 05-Sep-2024 10.0% +8 more	0 10.0% 20 10.0% +7 more	100 60.0% 50 40.0%
1	5001	3001	Midterm	2024-09-01	0	100
2	5002	3002	Midterm	03/09/2024	7	100
3	5003	3002	Final Assessment	14/09/2024	20	50
4	5004	3003	Midterm	05-Sep-2024	null	100
5	5005	3003	Final Assessment	16-Sep-2024	27	50
6	5006	3004	Midterm	2024-09-07	21	100
7	5007	3004	Final Assessment	2024-09-18	34	50
8	5008	3005	Midterm	09/09/2024	28	100
9	5009	3005	Final Assessment	20/09/2024	41	50
10	5010	3006	Midterm	11-Sep-2024	35	100

6. DATA CLEANING AND DATA TRANSFORMATION

- TRY_TO_NUMBER: Correct numeric types
- TRY_TO_DATE: Standardize date formats
- UPPER, INITCAP: Consistent naming
- TRIM: Remove spaces
- COALESCE: Manage missing data
- ROW_NUMBER: Eliminate repeated rows
- CASE
- WHERE
- INNER JOIN: Validate relationships

A. Student Table

QUALIFY ROW_NUMBER() OVER (...) = 1: Removes duplicate records for the same STUDENT_ID, ensuring only the first occurrence for each unique student is retained in the clean table.

```sql
CREATE OR REPLACE TABLE LAHARI_CLEAN_STUDENTS AS

SELECT
    TRY_TO_NUMBER(STUDENT_ID) AS STUDENT_ID,

    INITCAP(TRIM(STUDENT_NAME)) AS STUDENT_NAME,

    LOWER(TRIM(EMAIL)) AS EMAIL,

    UPPER(TRIM(DEPARTMENT)) AS DEPARTMENT,

    COALESCE(
        TRY_TO_DATE(ADMISSION_DATE, 'YYYY-MM-DD'),
        TRY_TO_DATE(ADMISSION_DATE, 'DD/MM/YYYY'),
        TRY_TO_DATE(ADMISSION_DATE, 'YYYY/MM/DD'),
        TRY_TO_DATE(ADMISSION_DATE, 'DD-MON-YYYY')
    ) AS ADMISSION_DATE,

    INITCAP(TRIM(CITY)) AS CITY,

    CASE
        WHEN EMAIL IS NULL OR TRIM(EMAIL) = ''
        THEN 'MISSING_EMAIL'
        ELSE 'VALID_EMAIL'
    END AS EMAIL_STATUS

FROM LAHARI_STUDENTS

WHERE TRY_TO_NUMBER(STUDENT_ID) IS NOT NULL

QUALIFY ROW_NUMBER() OVER (
    PARTITION BY STUDENT_ID
    ORDER BY STUDENT_ID
) = 1;
```

```sql
142  SELECT * FROM LAHARI_CLEAN_STUDENTS LIMIT 10;
```

esults (just now)

Table | Chart | Pivot — 10 rows — 353ms

	STUDENT_ID (1003 – 1030)	STUDENT_NAME (Aisha Khan 10.0%, Anvi Krishnan 10.0%, +8 more)	EMAIL (aarav.sharma@example.com 10.0%, aisha.khan@example.com 10.0%, +7 more)	DEPARTMENT (BUSINESS ANA... 30.0%, DATA SCIENCE 30.0%, +2 more)	ADMISSION_DATE (6/18/2024 – 8/3/2024)	CITY (Hyderabad 20.0%, Ahmedabad 10.0%, +7 more)	EMAIL_STATUS (VALID_EMAIL 90.0%, MISSING_EMAIL 10.0%)
1	1008	Saanvi Reddy	saanvi.reddy@example.com	DATA SCIENCE	2024-06-28	Hyderabad	VALID_EMAIL
2	1012	Ira Shah	aarav.sharma@example.com	INFORMATION TECHNOLO	2024-07-05	Ahmedabad	VALID_EMAIL
3	1017	Dhruv Mehta	dhruv.mehta@example.com	INFORMATION TECHNOLO	2024-07-13	Mumbai	VALID_EMAIL
4	1025	Ayaan Chawla	ayaan.chawla@example.com	BUSINESS ANALYTICS	2024-07-26	Delhi	VALID_EMAIL
5	1003	Ishaan Rao	ishaan.rao@example.com	DATA SCIENCE	2024-06-18	Bengaluru	VALID_EMAIL
6	1020	Sara Fernandes	sara.fernandes@example.com	BUSINESS ANALYTICS	2024-07-18	Goa	VALID_EMAIL
7	1004	Meera Nair	null	COMPUTER SCIENCE	2024-06-20	Kochi	MISSING_EMAIL
8	1028	Veda Mukherjee	veda.mukherjee@example.com	DATA SCIENCE	2024-07-31	Kolkata	VALID_EMAIL
9	1016	Anvi Krishnan	anvi.krishnan@example.com	COMPUTER SCIENCE	2024-07-12	Chennai	VALID_EMAIL
10	1030	Aisha Khan	aisha.khan@example.com	BUSINESS ANALYTICS	2024-08-03	Hyderabad	VALID_EMAIL

B. Course Table

```
CREATE OR REPLACE TABLE LAHARI_CLEAN_COURSES AS

SELECT
    TRY_TO_NUMBER(COURSE_ID) AS COURSE_ID,

    INITCAP(TRIM(COURSE_NAME)) AS COURSE_NAME,

    UPPER(TRIM(DEPARTMENT)) AS DEPARTMENT,

    TRIM(INSTRUCTOR_NAME) AS INSTRUCTOR_NAME,

    TRY_TO_NUMBER(COURSE_DURATION) AS COURSE_DURATION,

    TRY_TO_DECIMAL(COURSE_FEE, 10, 2) AS COURSE_FEE

FROM LAHARI_COURSES

WHERE TRY_TO_NUMBER(COURSE_ID) IS NOT NULL
  AND TRY_TO_NUMBER(COURSE_DURATION) > 0
  AND TRY_TO_DECIMAL(COURSE_FEE, 10, 2) > 0

QUALIFY ROW_NUMBER() OVER (
    PARTITION BY COURSE_ID
    ORDER BY COURSE_ID
) = 1;
```

```
174  SELECT * FROM LAHARI_CLEAN_COURSES;
```

Results (just now)

Table | Chart | Pivot — 10 rows — 201ms

	COURSE_ID	COURSE_NAME	DEPARTMENT	INSTRUCTOR_NAME	COURSE_DURATION	COURSE_FEE
	201 – 210	Business Intelligence 10.0% Cloud Fundamentals 10.0% +8 more	COMPUTER SCIENCE 30.0% BUSINESS ANALYTICS 20.0% +3 more	Dr. Kavitha Rao 10.0% Dr. Meenal Joshi 10.0% +8 more	6 – 14	5500 – 12000
1	210	Web Development	COMPUTER SCIENCE	Prof. Kiran Reddy	11	8800.00
2	208	Digital Electronics	ELECTRONICS	Prof. Sanjay Mehta	12	8200.00
3	205	Data Visualization	BUSINESS ANALYTICS	Dr. Sneha Iyer	6	5500.00
4	201	Python Programming	COMPUTER SCIENCE	Dr. Kavitha Rao	12	8500.00
5	209	Business Intelligence	BUSINESS ANALYTICS	Dr. Meenal Joshi	9	9500.00
6	204	Cloud Fundamentals	INFORMATION TECHNOLOGY	Prof. Rahul Shah	8	6500.00
7	203	Machine Learning	DATA SCIENCE	Dr. Nisha Menon	14	12000.00
8	202	Database Management	INFORMATION TECHNOLOGY	Prof. Arvind Kumar	10	7800.00
9	206	Cybersecurity Basics	COMPUTER SCIENCE	Prof. Vikram Das	10	9000.00
10	207	Statistics For Analytics	DATA SCIENCE	Dr. Pooja Nair	8	7000.00

C. Enrollments Table

- Removing exact duplicate enrollment records: DEDUPLICATED
- TRANSFORMED converts dates and numerical values.
- INNER JOIN ensures student and course references exist.
- CASE handles invalid progress percentages.
- Invalid enrollment dates are retained as NULL for auditing.

```sql
CREATE OR REPLACE TABLE LAHARI_CLEAN_ENROLLMENTS AS
WITH DEDUPLICATED AS (
    SELECT *
    FROM LAHARI_RAW_ENROLLMENTS
    QUALIFY ROW_NUMBER() OVER (
        PARTITION BY
            ENROLLMENT_ID,
            STUDENT_ID,
            COURSE_ID,
            ENROLLMENT_DATE,
            COMPLETION_STATUS,
            PROGRESS_PERCENTAGE
        ORDER BY ENROLLMENT_ID
    ) = 1
),
TRANSFORMED AS (
    SELECT
        TRY_TO_NUMBER(ENROLLMENT_ID) AS ENROLLMENT_ID,
        TRY_TO_NUMBER(STUDENT_ID) AS STUDENT_ID,
        TRY_TO_NUMBER(COURSE_ID) AS COURSE_ID,
        COALESCE(
            TRY_TO_DATE(ENROLLMENT_DATE, 'YYYY-MM-DD'),
            TRY_TO_DATE(ENROLLMENT_DATE, 'DD/MM/YYYY'),
            TRY_TO_DATE(ENROLLMENT_DATE, 'DD-MON-YYYY')
        ) AS ENROLLMENT_DATE,
        COALESCE(
            NULLIF(UPPER(TRIM(COMPLETION_STATUS)), ''),
            'UNKNOWN'
        ) AS COMPLETION_STATUS,
        CASE
            WHEN TRY_TO_NUMBER(PROGRESS_PERCENTAGE) BETWEEN 0 AND 100
            THEN TRY_TO_NUMBER(PROGRESS_PERCENTAGE)
            ELSE NULL
        END AS PROGRESS_PERCENTAGE,
        CASE
            WHEN TRY_TO_NUMBER(PROGRESS_PERCENTAGE) BETWEEN 0 AND 100
            THEN 'VALID'
            ELSE 'INVALID_PROGRESS'
        END AS PROGRESS_STATUS
    FROM DEDUPLICATED

)
SELECT T.*
FROM TRANSFORMED T
INNER JOIN LAHARI_CLEAN_STUDENTS S
    ON T.STUDENT_ID = S.STUDENT_ID
INNER JOIN LAHARI_CLEAN_COURSES C
    ON T.COURSE_ID = C.COURSE_ID
WHERE T.ENROLLMENT_ID IS NOT NULL;
```

Results (just now)

Table | Chart | Pivot — 10 rows — 332ms

	ENROLLMENT_ID (3002 – 3052)	STUDENT_ID (1001 – 1026)	COURSE_ID (201 – 209)	ENROLLMENT_DATE (8/8/2024 – 10/22/2024)	COMPLETION_STATUS (NOT_STARTED 40.0%, COMPLETED 30.0%, +1 more)	PROGRESS_PERCENTAGE (0 – 100)	PROGRESS_STATUS (VALID 90.0%, INVALID_PROGRESS 10.0%)
1	3045	1023	203	2024-10-06	IN_PROGRESS	65	VALID
2	3031	1016	206	2024-09-15	COMPLETED	100	VALID
3	3049	1025	205	2024-10-12	COMPLETED	100	VALID
4	3017	1009	209	2024-08-25	NOT_STARTED	null	INVALID_PROGRESS
5	3028	1014	202	2024-09-16	NOT_STARTED	0	VALID
6	3034	1017	201	2024-09-25	NOT_STARTED	0	VALID
7	3052	1026	208	2024-10-22	NOT_STARTED	0	VALID
8	3018	1009	207	2024-09-01	COMPLETED	100	VALID
9	3002	1001	203	2024-08-08	IN_PROGRESS	65	VALID
10	3027	1014	204	2024-09-09	IN_PROGRESS	65	VALID

D. Assessment Table

- Identify missing marks.
- Remove invalid marks and marks exceeding total marks.
- Exclude assessments referencing invalid enrollments.

```sql
CREATE OR REPLACE TABLE LAHARI_CLEAN_ASSESSMENTS AS
WITH TRANSFORMED AS (
    SELECT
        TRY_TO_NUMBER(ASSESSMENT_ID) AS ASSESSMENT_ID,
        TRY_TO_NUMBER(ENROLLMENT_ID) AS ENROLLMENT_ID,
        INITCAP(TRIM(ASSESSMENT_NAME)) AS ASSESSMENT_NAME,
        COALESCE(
            TRY_TO_DATE(ASSESSMENT_DATE, 'YYYY-MM-DD'),
            TRY_TO_DATE(ASSESSMENT_DATE, 'DD/MM/YYYY'),
            TRY_TO_DATE(ASSESSMENT_DATE, 'DD-MON-YYYY')
        ) AS ASSESSMENT_DATE,
        TRY_TO_NUMBER(NULLIF(TRIM(MARKS_OBTAINED), '')) AS MARKS_OBTAINED,
        TRY_TO_NUMBER(TOTAL_MARKS) AS TOTAL_MARKS
    FROM LAHARI_ASSESSMENTS
)
SELECT
    A.*,
    CASE
        WHEN A.MARKS_OBTAINED IS NULL
        THEN 'MISSING_MARKS'
        ELSE 'VALID_MARKS'
    END AS MARKS_STATUS,
    CASE
        WHEN A.MARKS_OBTAINED IS NOT NULL
             AND A.TOTAL_MARKS > 0
        THEN ROUND(
            (A.MARKS_OBTAINED / A.TOTAL_MARKS) * 100,
            2
        )
        ELSE NULL
    END AS SCORE_PERCENTAGE,
    CASE
        WHEN A.MARKS_OBTAINED IS NULL THEN 'NOT_EVALUATED'
        WHEN A.MARKS_OBTAINED / NULLIF(A.TOTAL_MARKS, 0) >= 0.40
            THEN 'PASS'
        ELSE 'FAIL'
    END AS RESULT_STATUS
FROM TRANSFORMED A
INNER JOIN LAHARI_CLEAN_ENROLLMENTS E
    ON A.ENROLLMENT_ID = E.ENROLLMENT_ID
WHERE A.ASSESSMENT_ID IS NOT NULL
  AND A.TOTAL_MARKS > 0
  AND (
      A.MARKS_OBTAINED IS NULL
      OR A.MARKS_OBTAINED BETWEEN 0 AND A.TOTAL_MARKS
  )
QUALIFY ROW_NUMBER() OVER (
    PARTITION BY A.ASSESSMENT_ID
    ORDER BY A.ASSESSMENT_ID
) = 1;
```

Results (just now)

Table | Chart | Pivot — 10 rows — 175ms

	ASSESSMENT_ID (5001–5010)	ENROLLMENT_ID (3001–3006)	ASSESSMENT_NAME (Midterm 60.0%, Final Assessment 40.0%)	ASSESSMENT_DATE (9/1/2024–9/20/2024)	MARKS_OBTAINED (0–41)	TOTAL_MARKS (50–100)	MARKS_STATUS (VALID_MARKS 90.0%, MISSING_MA... 10.0%)	SCORE_PERCENTAGE (0–82)	RESULT_STATUS (FAIL 50.0%, PASS 40.0%, +1 more)
1	5001	3001	Midterm	2024-09-01	0	100	VALID_MARKS	0.00	FAIL
2	5002	3002	Midterm	2024-09-03	7	100	VALID_MARKS	7.00	FAIL
3	5003	3002	Final Assessment	2024-09-14	20	50	VALID_MARKS	40.00	PASS
4	5004	3003	Midterm	2024-09-05	null	100	MISSING_MARKS	null	NOT_EVALUATED
5	5005	3003	Final Assessment	2024-09-16	27	50	VALID_MARKS	54.00	PASS
6	5006	3004	Midterm	2024-09-07	21	100	VALID_MARKS	21.00	FAIL
7	5007	3004	Final Assessment	2024-09-18	34	50	VALID_MARKS	68.00	PASS
8	5008	3005	Midterm	2024-09-09	28	100	VALID_MARKS	28.00	FAIL
9	5009	3005	Final Assessment	2024-09-20	41	50	VALID_MARKS	82.00	PASS
10	5010	3006	Midterm	2024-09-11	35	100	VALID_MARKS	35.00	FAIL

7. ANALYTICAL SQL QUERIES

I. Department-wise student distribution

```sql
SELECT
    DEPARTMENT,
    COUNT(*) AS TOTAL_STUDENTS
FROM LAHARI_CLEAN_STUDENTS
GROUP BY DEPARTMENT
ORDER BY TOTAL_STUDENTS DESC;
```

sults (just now)

Table | Chart | Pivot — 5 rows — 665ms

	DEPARTMENT	TOTAL_STUDENTS
1	COMPUTER SCIENCE	7
2	DATA SCIENCE	6
3	INFORMATION TECHNOLOGY	6
4	ELECTRONICS	6
5	BUSINESS ANALYTICS	5

II. Course-wise enrollments

```sql
SELECT
    C.COURSE_ID,
    C.COURSE_NAME,
    C.INSTRUCTOR_NAME,
    COUNT(E.ENROLLMENT_ID) AS TOTAL_ENROLLMENTS
FROM LAHARI_CLEAN_COURSES C
INNER JOIN LAHARI_CLEAN_ENROLLMENTS E
    ON C.COURSE_ID = E.COURSE_ID
GROUP BY
    C.COURSE_ID,
    C.COURSE_NAME,
    C.INSTRUCTOR_NAME
ORDER BY TOTAL_ENROLLMENTS DESC;
```

Results (1 minute ago)

Table | Chart | Pivot — 10 rows — 431ms

	COURSE_ID (201 – 210)	COURSE_NAME (Business Intelligence 10.0%, Cloud Fundamentals 10.0%, +8 more)	INSTRUCTOR_NAME (Dr. Kavitha Rao 10.0%, Dr. Meenal Joshi 10.0%, +8 more)	TOTAL_ENROLLMENTS (5 – 6)
1	203	Machine Learning	Dr. Nisha Menon	6
2	206	Cybersecurity Basics	Prof. Vikram Das	6
3	205	Data Visualization	Dr. Sneha Iyer	6
4	209	Business Intelligence	Dr. Meenal Joshi	6
5	201	Python Programming	Dr. Kavitha Rao	6
6	208	Digital Electronics	Prof. Sanjay Mehta	6
7	207	Statistics For Analytics	Dr. Pooja Nair	6
8	204	Cloud Fundamentals	Prof. Rahul Shah	6
9	210	Web Development	Prof. Kiran Reddy	6
10	202	Database Management	Prof. Arvind Kumar	5

III. Average assessment score per course

```sql
SELECT
    C.COURSE_NAME,
    COUNT(A.ASSESSMENT_ID) AS TOTAL_ASSESSMENTS,
    ROUND(AVG(A.SCORE_PERCENTAGE), 2) AS AVG_SCORE,
    MAX(A.SCORE_PERCENTAGE) AS HIGHEST_SCORE,
    MIN(A.SCORE_PERCENTAGE) AS LOWEST_SCORE
FROM LAHARI_CLEAN_COURSES C

INNER JOIN LAHARI_CLEAN_ENROLLMENTS E
    ON C.COURSE_ID = E.COURSE_ID

INNER JOIN LAHARI_CLEAN_ASSESSMENTS A
    ON E.ENROLLMENT_ID = A.ENROLLMENT_ID

WHERE A.SCORE_PERCENTAGE IS NOT NULL

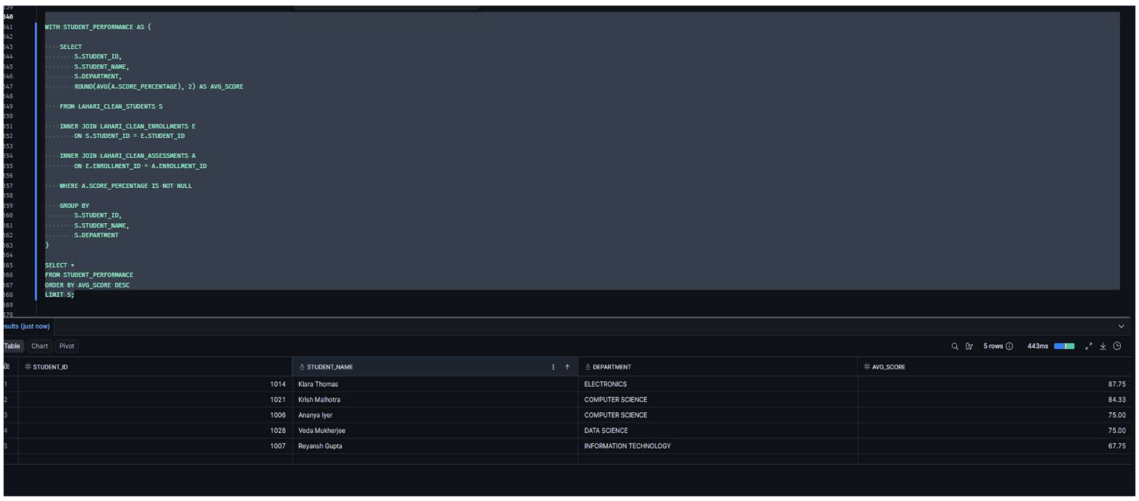
GROUP BY C.COURSE_NAME
ORDER BY AVG_SCORE DESC;
```

Results (just now)

Table | Chart | Pivot — 10 rows — 660ms

	COURSE_NAME (Business Intelligence 10.0%, Cloud Fundamentals 10.0%, +8 more)	TOTAL_ASSESSMENTS (7 – 12)	AVG_SCORE (35.43 – 58)	HIGHEST_SCORE (74 – 100)	LOWEST_SCORE (0 – 25)
1	Web Development	8	58.00	88.00	25.00
2	Digital Electronics	12	57.58	98.00	15.00
3	Database Management	9	54.44	98.00	22.00
4	Statistics For Analytics	9	53.00	92.00	18.00
5	Data Visualization	11	49.45	100.00	1.00
6	Python Programming	9	49.22	91.00	0.00
7	Cybersecurity Basics	9	47.89	99.00	8.00
8	Cloud Fundamentals	9	46.78	84.00	4.00
9	Machine Learning	12	45.83	90.00	5.00
10	Business Intelligence	7	35.43	74.00	11.00

IV. Top 5 performing students



```
WITH STUDENT_PERFORMANCE AS (

    SELECT
        S.STUDENT_ID,
        S.STUDENT_NAME,
        S.DEPARTMENT,
        ROUND(AVG(A.SCORE_PERCENTAGE), 2) AS AVG_SCORE

    FROM LAHARI_CLEAN_STUDENTS S

    INNER JOIN LAHARI_CLEAN_ENROLLMENTS E
        ON S.STUDENT_ID = E.STUDENT_ID

    INNER JOIN LAHARI_CLEAN_ASSESSMENTS A
        ON E.ENROLLMENT_ID = A.ENROLLMENT_ID

    WHERE A.SCORE_PERCENTAGE IS NOT NULL

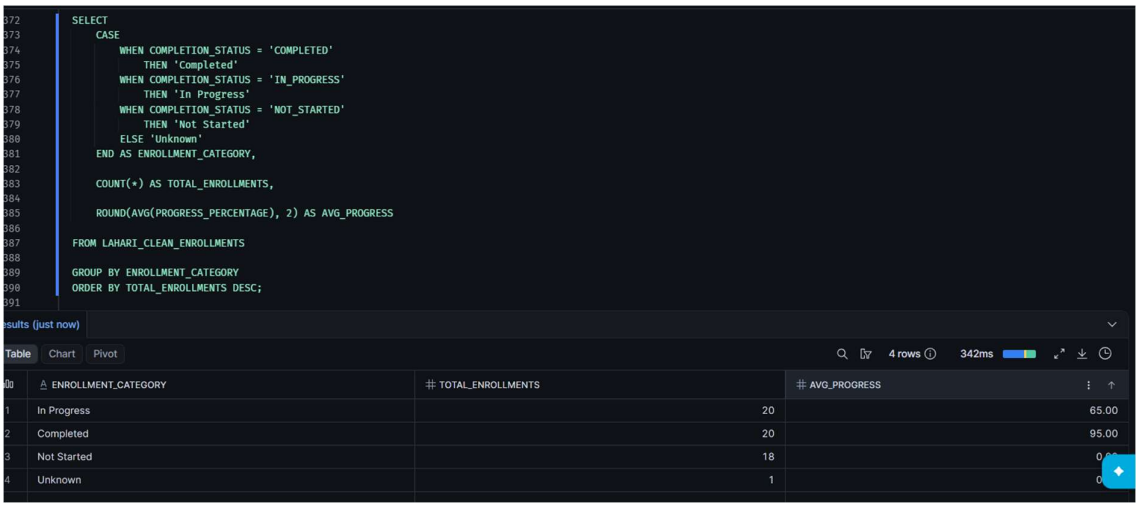
    GROUP BY
        S.STUDENT_ID,
        S.STUDENT_NAME,
        S.DEPARTMENT
)

SELECT *
FROM STUDENT_PERFORMANCE
ORDER BY AVG_SCORE DESC
LIMIT 5;
```

Results (just now) — Table | Chart | Pivot — 5 rows — 443ms

	STUDENT_ID	STUDENT_NAME	DEPARTMENT	AVG_SCORE
1	1014	Klara Thomas	ELECTRONICS	87.75
2	1021	Krish Malhotra	COMPUTER SCIENCE	84.33
3	1006	Ananya Iyer	COMPUTER SCIENCE	75.00
4	1028	Veda Mukherjee	DATA SCIENCE	75.00
5	1007	Reyansh Gupta	INFORMATION TECHNOLOGY	67.75

V. Completed vs on going enrollments



```
SELECT
    CASE
        WHEN COMPLETION_STATUS = 'COMPLETED'
            THEN 'Completed'
        WHEN COMPLETION_STATUS = 'IN_PROGRESS'
            THEN 'In Progress'
        WHEN COMPLETION_STATUS = 'NOT_STARTED'
            THEN 'Not Started'
        ELSE 'Unknown'
    END AS ENROLLMENT_CATEGORY,

    COUNT(*) AS TOTAL_ENROLLMENTS,

    ROUND(AVG(PROGRESS_PERCENTAGE), 2) AS AVG_PROGRESS

FROM LAHARI_CLEAN_ENROLLMENTS

GROUP BY ENROLLMENT_CATEGORY
ORDER BY TOTAL_ENROLLMENTS DESC;
```

Results (just now) — Table | Chart | Pivot — 4 rows — 342ms

	ENROLLMENT_CATEGORY	TOTAL_ENROLLMENTS	AVG_PROGRESS
1	In Progress	20	65.00
2	Completed	20	95.00
3	Not Started	18	0.00
4	Unknown	1	0

VI. Instructor-wise student enrollment

```sql
SELECT
    C.INSTRUCTOR_NAME,
    COUNT(E.ENROLLMENT_ID) AS TOTAL_STUDENTS,
    COUNT(DISTINCT C.COURSE_ID) AS TOTAL_COURSES
FROM LAHARI_CLEAN_COURSES C

INNER JOIN LAHARI_CLEAN_ENROLLMENTS E
    ON C.COURSE_ID = E.COURSE_ID

GROUP BY C.INSTRUCTOR_NAME

HAVING COUNT(E.ENROLLMENT_ID) >= 5

ORDER BY TOTAL_STUDENTS DESC;
```

Add to Chat | Explain | Quick edit | Format

Table | Chart | Pivot — 10 rows — 460ms

	INSTRUCTOR_NAME	TOTAL_STUDENTS	TOTAL_COURSES
	Dr. Kavitha Rao 10.0% Dr. Meenal Joshi 10.0% +8 more	5 — 6	1 — 1
1	Dr. Nisha Menon	6	1
2	Prof. Vikram Das	6	1
3	Dr. Sneha Iyer	6	1
4	Dr. Meenal Joshi	6	1
5	Prof. Sanjay Mehta	6	1
6	Dr. Pooja Nair	6	1
7	Prof. Rahul Shah	6	1
8	Prof. Kiran Reddy	6	1

VII. Courses with average score below 60%

```sql
SELECT
    C.COURSE_NAME,
    ROUND(AVG(A.SCORE_PERCENTAGE), 2) AS AVG_SCORE,
    COUNT(A.ASSESSMENT_ID) AS ASSESSMENT_COUNT
FROM LAHARI_CLEAN_COURSES C

INNER JOIN LAHARI_CLEAN_ENROLLMENTS E
    ON C.COURSE_ID = E.COURSE_ID

INNER JOIN LAHARI_CLEAN_ASSESSMENTS A
    ON E.ENROLLMENT_ID = A.ENROLLMENT_ID

WHERE A.SCORE_PERCENTAGE IS NOT NULL
GROUP BY C.COURSE_NAME
HAVING AVG(A.SCORE_PERCENTAGE) < 60
ORDER BY AVG_SCORE;
```

Results (just now)

Table | Chart | Pivot — 10 rows — 601ms

	COURSE_NAME	AVG_SCORE	ASSESSMENT_COUNT
	Business Intelligence 10.0% Cloud Fundamentals 10.0% +8 more	35.43 — 58	7 — 12
1	Business Intelligence	35.43	7
2	Machine Learning	45.83	12
3	Cloud Fundamentals	46.78	9
4	Cybersecurity Basics	47.89	9
5	Python Programming	49.22	9
6	Data Visualization	49.45	11

VIII. Department-wise academic performance

```sql
SELECT
    S.DEPARTMENT,
    COUNT(DISTINCT S.STUDENT_ID) AS TOTAL_STUDENTS,
    COUNT(A.ASSESSMENT_ID) AS TOTAL_ASSESSMENTS,
    ROUND(AVG(A.SCORE_PERCENTAGE), 2) AS AVG_DEPARTMENT_SCORE,
    SUM(CASE
        WHEN A.RESULT_STATUS = 'PASS' THEN 1
        ELSE 0
    END) AS TOTAL_PASS,
    SUM(CASE
        WHEN A.RESULT_STATUS = 'FAIL' THEN 1
        ELSE 0
    END) AS TOTAL_FAIL
FROM LAHARI_CLEAN_STUDENTS S
INNER JOIN LAHARI_CLEAN_ENROLLMENTS E
    ON S.STUDENT_ID = E.STUDENT_ID
INNER JOIN LAHARI_CLEAN_ASSESSMENTS A
    ON E.ENROLLMENT_ID = A.ENROLLMENT_ID
GROUP BY S.DEPARTMENT
ORDER BY AVG_DEPARTMENT_SCORE DESC;
```

Results (just now)

Table | Chart | Pivot — 5 rows — 581ms

	DEPARTMENT	TOTAL_STUDENTS	TOTAL_ASSESSMENTS	AVG_DEPARTMENT_SCORE	TOTAL_PASS	TOTAL_FAIL
1	INFORMATION TECHNOLOGY	6	22	53.90	15	6
2	BUSINESS ANALYTICS	4	16	53.31	10	6
3	ELECTRONICS	5	18	51.11	11	
4	DATA SCIENCE	6	18	50.39	10	
5	COMPUTER SCIENCE	7	22	43.05	13	9

IX. Students who have not completed courses

```sql
SELECT
    S.STUDENT_NAME,
    S.DEPARTMENT,
    C.COURSE_NAME,
    E.PROGRESS_PERCENTAGE,
    E.COMPLETION_STATUS
FROM LAHARI_CLEAN_STUDENTS S

INNER JOIN LAHARI_CLEAN_ENROLLMENTS E
    ON S.STUDENT_ID = E.STUDENT_ID

INNER JOIN LAHARI_CLEAN_COURSES C
    ON E.COURSE_ID = C.COURSE_ID

WHERE E.COMPLETION_STATUS <> 'COMPLETED'

ORDER BY E.PROGRESS_PERCENTAGE ASC NULLS FIRST;
```

Results (just now)

Table | Chart | Pivot — 39 rows — 772ms

	STUDENT_NAME	DEPARTMENT	COURSE_NAME	PROGRESS_PERCENTAGE	COMPLETION_STATUS
	Arjun Kumar 5.1% Dhruv Mehta 5.1% +27 more	COMPUTER SCIENCE 23.1% ELECTRONICS 23.1% +3 more	Business Intelligence 10.3% Cloud Fundamentals 10.3% +8 more	0 — 65	IN_PROGRESS 51.3% NOT_STARTED 46.2% +1 more
1	Vivaan Das	ELECTRONICS	Business Intelligence	null	NOT_STARTED
2	Kabir Singh	ELECTRONICS	Data Visualization	null	NOT_STARTED
3	Krish Malhotra	COMPUTER SCIENCE	Python Programming	0	NOT_STARTED
4	Kiara Thomas	ELECTRONICS	Database Management	0	NOT_STARTED
5	Dhruv Mehta	INFORMATION TECHNOLOGY	Python Programming	0	NOT_STARTED
6	Tara Sinha	COMPUTER SCIENCE	Digital Electronics	0	NOT_STARTED

X. Rank courses by enrollment

```sql
WITH COURSE_ENROLLMENT AS (
    SELECT
        C.COURSE_ID,
        C.COURSE_NAME,
        C.DEPARTMENT,
        COUNT(E.ENROLLMENT_ID) AS TOTAL_ENROLLMENTS
    FROM LAHARI_CLEAN_COURSES C
    LEFT JOIN LAHARI_CLEAN_ENROLLMENTS E
        ON C.COURSE_ID = E.COURSE_ID
    GROUP BY
        C.COURSE_ID,
        C.COURSE_NAME,
        C.DEPARTMENT
)
SELECT
    COURSE_NAME,
    DEPARTMENT,
    TOTAL_ENROLLMENTS,
    RANK() OVER (
        ORDER BY TOTAL_ENROLLMENTS DESC
    ) AS COURSE_RANK
FROM COURSE_ENROLLMENT
ORDER BY COURSE_RANK;
```

Chart Pivot 10 rows 445ms

COURSE_NAME	DEPARTMENT	TOTAL_ENROLLMENTS	COURSE_RANK
Business Intelligence 10.0% Cloud Fundamentals 10.0% +8 more	COMPUTER SCIENCE 30.0% BUSINESS ANALYTICS 20.0% +3 more	5 – 6	1 – 10
Machine Learning	DATA SCIENCE	6	1
Cybersecurity Basics	COMPUTER SCIENCE	6	1
Data Visualization	BUSINESS ANALYTICS	6	1
Business Intelligence	BUSINESS ANALYTICS	6	1
Python Programming	COMPUTER SCIENCE	6	1

8. QUERY PROFILE

